Causal Inference with Panel Data

A short course in six lectures

Yiqing Xu, Stanford University
First taught at Waseda University, 2026

Overview

Panel data are the most common basis for causal claims in the social sciences. For decades the default analysis was a two-way fixed effects regression. Over the past ten years that default has been challenged, and dozens of alternative estimators have appeared. This short course works through the change, from the classical models to the methods now in use.

The first half covers the parametric panel models and what their assumptions require, then difference-in-differences (DID) and factorial DID as distinctive research designs. It closes with the result that two-way fixed effects recovers a weighted average of treatment effects whose weights can turn negative when units are treated at different times, as well as other pathologies. The second half covers the responses: heterogeneity-robust DID estimators, the synthetic control method and latent factor models, as well as doubly and triply robust methods.

Each lecture pairs the identification argument with applied examples and diagnostics: what to plot, what to test, and what to report when the assumptions are in doubt.

Audience and prerequisites

The course is for applied researchers and graduate students in the social sciences who work with panel data. Participants should have taken a graduate or advanced-undergraduate regression course. Prior experience with R is helpful but not required. Each method is introduced through an example and a picture, and formulas appear only where the ideas need them.

Format and materials

Six lectures of 100 to 120 minutes each. The slides, their LaTeX sources, the R scripts that regenerate every figure, and the bibliography are free to use and adapt:

- Course site: https://teaching.yiqingxu.org/panel-lectures/
- Source repository: https://github.com/xuyiqing/panel-lectures

Implementation is in R, mainly with `panelView`, `fect` and `fdid`; other packages (`did`, `bacondecomp`, `augsynth`, `gsynth`) appear where a lecture uses them.

For an overview of the field and of the approach taken here:

Xu, Yiqing. 2023. "Causal Inference with Time-Series Cross-Sectional Data: A Reflection." In *The Oxford Handbook of Engaged Methodological Pluralism in Political Science,* 353–74. Oxford University Press.

Schedule and readings

Lecture 1. The parametric approach

Panel setup and unobserved confounding; fixed effects regression; modeling time; modeling dynamic and heterogeneous effects.

Core readings

Hainmueller, Jens, and Dominik Hangartner. 2019. "Does Direct Democracy Hurt Immigrant Minorities? Evidence from Naturalization Decisions in Switzerland." *American Journal of Political Science* 63 (3): 530–47.

Stock, James H., and Mark W. Watson. 2007. *Introduction to Econometrics.* 2nd ed. Pearson Addison Wesley.

Further readings

Holland, Paul W. 1986. "Statistics and Causal Inference." *Journal of the American Statistical Association* 81 (396): 945–60.

Lecture 2. Difference-in-differences

Motivating examples; setup and identification; estimation; threats to validity.

Core readings

Abadie, Alberto. 2005. "Semiparametric Difference-in-Differences Estimators." *Review of Economic Studies* 72 (1): 1–19.

Card, David, and Alan B. Krueger. 1994. "Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania." *American Economic Review* 84 (4): 772–93.

Further readings

Roth, Jonathan, and Pedro H. C. Sant'Anna. 2023. "When Is Parallel Trends Sensitive to Functional Form?" *Econometrica* 91 (2): 737–47.

Lecture 3. Factorial difference-in-differences

Events that affect every unit; setup and estimands; identification; extensions; application to clans and the Great Famine.

Core readings

Xu, Yiqing, Anqi Zhao, and Peng Ding. 2026. "Factorial Difference-in-Differences." *Journal of the American Statistical Association* 121 (554): 963–75.

Further readings

Cao, Jiarui, Yiqing Xu, and Chuanchuan Zhang. 2022. "Clans and Calamity: How Social Capital Saved Lives during China's Great Famine." *Journal of Development Economics* 157:102865.

VanderWeele, Tyler J. 2009. "On the Distinction between Interaction and Effect Modification." *Epidemiology* 20 (6): 863–71.

Lecture 4. Two-way fixed effects revisited

Two-way fixed effects and its assumptions; the design problem; parallel trends violations; the weighting problem under heterogeneous effects.

Core readings

Goodman-Bacon, Andrew. 2021. "Difference-in-Differences with Variation in Treatment Timing." *Journal of Econometrics* 225 (2): 254–77.

Imai, Kosuke, and In Song Kim. 2021. "On the Use of Two-Way Fixed Effects Regression Models for Causal Inference with Panel Data." *Political Analysis* 29 (3): 405–15.

Further readings

de Chaisemartin, Clément, and Xavier D'Haultfœuille. 2020. "Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects." *American Economic Review* 110 (9): 2964–96.

Lecture 5. Modern difference-in-differences

Heterogeneity-robust estimators; diagnostics; lessons from 49 reanalyses of published studies.

Core readings

Chiu, Albert, Xingchen Lan, Ziyi Liu, and Yiqing Xu. 2026. "Causal Panel Analysis under Parallel Trends: Lessons from a Large Reanalysis Study." *American Political Science Review* 120 (1): 245–66.

Liu, Licheng, Ye Wang, and Yiqing Xu. 2024. "A Practical Guide to Counterfactual Estimators for Causal Inference with Time-Series Cross-Sectional Data." *American Journal of Political Science* 68 (1): 160–76.

Further readings

Borusyak, Kirill, Xavier Jaravel, and Jann Spiess. 2024. "Revisiting Event-Study Designs: Robust and Efficient Estimation." *Review of Economic Studies* 91 (6): 3253–85.

Callaway, Brantly, and Pedro H. C. Sant'Anna. 2021. "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics* 225 (2): 200–230.

Rambachan, Ashesh, and Jonathan Roth. 2023. "A More Credible Approach to Parallel Trends." *Review of Economic Studies* 90 (5): 2555–91.

Sun, Liyang, and Sarah Abraham. 2021. "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects." *Journal of Econometrics* 225 (2): 175–99.

Lecture 6. Synthetic control and extensions

The synthetic control method; the latent factor approach; doubly and triply robust methods.

Core readings

Abadie, Alberto, Alexis Diamond, and Jens Hainmueller. 2010. "Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California's Tobacco Control Program." *Journal of the American Statistical Association* 105 (490): 493–505.

Xu, Yiqing. 2017. "Generalized Synthetic Control Method: Causal Inference with Interactive Fixed Effects Models." *Political Analysis* 25 (1): 57–76.

Further readings

Arkhangelsky, Dmitry, Susan Athey, David A. Hirshberg, Guido W. Imbens, and Stefan Wager. 2021. "Synthetic Difference-in-Differences." *American Economic Review* 111 (12): 4088–118.

Athey, Susan, Mohsen Bayati, Nikolay Doudchenko, Guido Imbens, and Khashayar Khosravi. 2021. "Matrix Completion Methods for Causal Panel Data Models." *Journal of the American Statistical Association* 116 (536): 1716–30.

Athey, Susan, Guido Imbens, Zhaonan Qu, and Davide Viviano. 2026. "Triply Robust Panel Estimators." *Journal of Applied Econometrics* 41 (6): 745–60.

Ben-Michael, Eli, Avi Feller, and Jesse Rothstein. 2021. "The Augmented Synthetic Control Method." *Journal of the American Statistical Association* 116 (536): 1789–803.